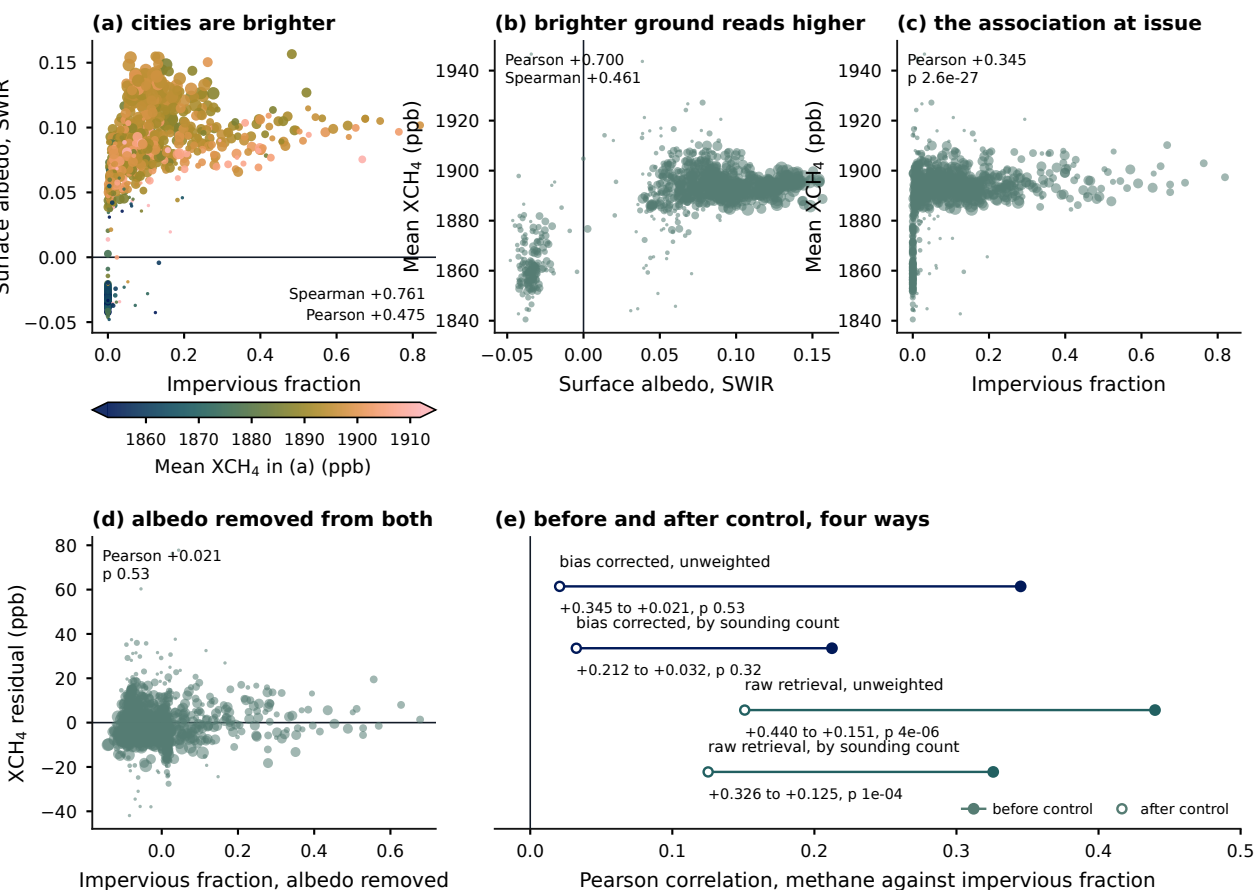




Every number here is recomputed at build time from `analysis_grid_2018.csv` and `methane_covariates_2018.csv`, and the figure refuses to build unless the bias-corrected values reproduce `albedo_confounder_2018.csv`. Mark area is the sounding count on the composite figure's half-decade classes: a cell's value is a mean over 1 to 410 soundings and the correlations differ between weighted and unweighted, so (e) reports both. 166 of the 926 cells carry a negative mean SWIR albedo. That is not an error and not a fill value: albedo here is a parameter the retrieval fits, not a reflectance it measures, and over dark surfaces the fit lands below zero; see `data/processed/README.md` for why those cells are kept. This does not show that the urban signal is an artefact. At Spearman +0.76 there is not enough independent variation to say which of the two is doing the work, and cities really are brighter, so controlling for albedo over-controls by an unknown amount. Nor is the partial correlation the corrected estimate: on the raw retrieval it survives at +0.15, and that field carries the larger uncorrected bias, so incomplete control reads as well as a real signal. Separating them needs a retrieval known to be albedo-unbiased, or urban extent varying at constant albedo. This region provides neither, which is a limit of the study design rather than of the analysis. See `ERRATA.md 7.4`.